

Scalable Conjunction Risk Assessment and Maneuver Timing Optimization for Low Earth Orbit (LEO) Satellite

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Abstract:

The growing amount of objects in Low Earth Orbit (LEO) has created an increase in difficulty for managing operational storage related to conjunction evaluation and collision avoidance. The current study develops a scalable and deterministic system for conjunction risk screening and maneuver optimization including precise time of closest approach (TCA), three-dimensional covariance-based collision probability, and machine learning-aided ΔV estimation. Epochs are generated via deterministic orbital propagation using SGP4 state generation. A two-step detection process uses coarse discretized screening followed by quadratic refinement of the square of the relative distance to generate sub-second accuracy of TCA. Three-dimensional Gaussian formulations of collision probability are utilized as well as the RTN frame. The encounter scenarios were generated analytically and used as input data for a regression-type machine learning model to predict optimal values of ΔV to use when performing high-speed screening over large numbers of debris. Benchmark testing shows almost linear performance improvement in terms of the number of pieces of debris assessed. Testing also demonstrated improved TCA and reduced ΔV usage in comparison to other rough sampling techniques. The structure developed does have limitations associated with it, specifically the assumptions regarding diagonal covariances and impulsive burns may not always yield sufficient or realistic results; however, this methodology could serve as a basis for developing more robust and scalable systems for conjunction assessments and collision avoidance.

Keywords: Low Earth Orbit, Conjunction Assessment, Collision Probability, Covariance Modelling, Maneuver Optimization, ΔV Regression, Autonomous Screening.

I. Introduction

The number of man-made objects being placed into low earth orbit (LEO), has resulted in a much greater rate of space object close approach events. [2] Current Conjunction Assessment processes utilize deterministic orbital propagators and use covariance-based collision probability estimations which often require considerable computer processing time and resources when evaluating large

collections of debris. [3] As satellite constellations continue to grow, there is an increasing need for scalable means of risk assessment and maneuver optimization. [2][4]. Historically, the collision probability was estimated using traditional methods based upon the use of Gaussian covariance models within the encounter plane. [4] The operational systems utilizing covariance propagation, high fidelity orbit determination has been shown to be very computationally intensive for large scale autonomous screenings. In addition to the dangers they pose to operational satellites these fragments can also collide with each other creating even more debris. [5]

In this paper we propose a modular and scalable framework that combines deterministic orbital propagation with 3-Dimensional covariance-based collision probability modelling, Refinement of Time Criticality Analysis (TCA), and constrained maneuver optimization. We do not use machine learning as a decision authority. However, we do use machine learning as a computational accelerator for calculating ΔV required during large scale encounter evaluations. This could threaten the future of space missions since this could render space unusable. Space collisions are not only economically harmful to space agencies but are also a significant threat to the safe and sustainable operation of space. If no space debris is removed, in the long term, debris would result in loss of 1.95 percent of worldwide gross domestic product (GDP).[6] Currently, the largest danger to space collisions is posed by un-tracked space debris, although the problem is expected to shift towards tracked objects as satellite-to-satellite collisions become more prevalent in the next few decades as larger numbers of satellites enter LEO.

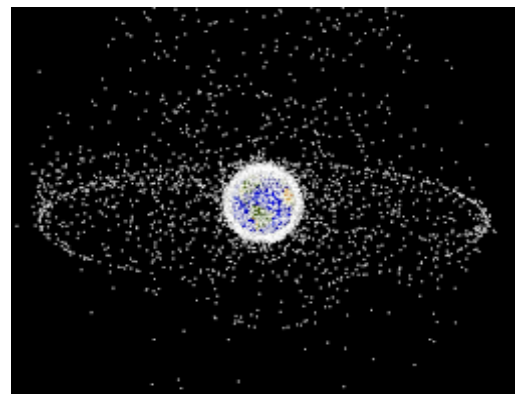


Figure 1 Debris Representation
source: SpaceNews.com

The current methods employed for assessing collision risks include the use of Conjunction Data Messages (CDMs) as well as associated data. These messages can provide relevant information regarding an object-to-object conjunction including both position and velocity vectors at the closest point of approach; also included are covariances that represent uncertainties. In addition to evaluating collision risks, satellite owners and operator's use this information to determine whether a manoeuvre should be performed, either prior to or after receipt of a collision alert from their respective agencies or organizations [22]. Important parameters to consider when determining whether to perform a manoeuvre include the probability of collision (PC); miss-distance; time to the time of closest approach (TCA); relative velocity; radial separation; position uncertainty; and collision geometry [23]. It has been found that obtaining accurate and timely assessments of collision risk/probability-of-collision is essential for implementing Collision Avoidance (CA) strategies. When the probability of collision is accurately estimated, it allows for fewer manoeuvres than necessary, while simultaneously reducing overall aggregate collision risk across operational fleets. However, there are many factors that contribute to estimating collision risk/probability-of-collision. Therefore, there are numerous methods used to estimate the probability of collision between satellites and/or between satellites and space debris. Due to increasing reliance on artificial intelligence/machine learning technology, in future years, it is expected that most space operators will automate the process of planning manoeuvres in response to receiving collision warnings [22], [14].

The major contributions made within this research are:

1. A two-step method to detect TCA using quadratic refinement of relative distances.
2. A three-dimensional RTN frame collision probability model using covariance.
3. Optimizing burn lead times using minimum Δv with separation constraint.
4. Rapid maneuver estimation through ML-assisted Δv regression.
5. Scalable performance assessment over variable sizes of debris populations.

II. Methodology

The ability to track, determine positions and predict orbits of objects in space as well as identify possible close approaches between these objects in space is critical to creating a safe and sustainable environment for conducting space operations. Space traffic management provides both regulatory and technical solutions to ensure that operators have access to space safely. While there are many ways to accomplish this, the basic pipeline for avoiding collisions is

very much alike. It typically starts at identifying objects you wish to track using Space-Track.

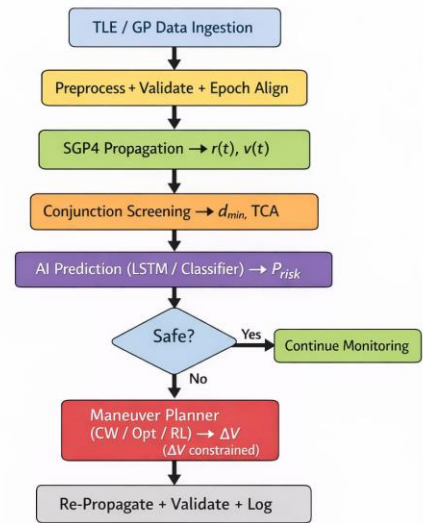


Figure 2 Methodology

Once tracked, their orbital paths can be determined or predicted to see if they will ever cross each other. If they may potentially cross each other, then the next step would be to perform a “conjunction screen.” This type of screen is generally performed on a one-to-many basis on one object against every other object, to determine if there exists a potential close approach between any of the space objects. Based upon the results of the conjunction screen, those identified as being high risk may result in additional assessments of collision risk. These assessments include evaluating the probability of collision and the impact of that collision, as well as the overall consequences associated with such an event. As previously stated, the collision risk is simply the product of the probability of collision and the impact/consequences of such an event [8].

In addition, for those conjunctions identified as being a high risk, and after assessing the collision risk; and determining that there is indeed a need to maneuver away from a potential collision, then alerts in the form of Conjunction Data Messages (CDM's) are generated. These messages are continuously updated until the time of closest approach (TCA) as new information becomes available.

II.1 Orbital Propagation and Conjunction Screening

Orbital states are generated using the SGP4 propagator with deterministic epoch control [1] [2]. It predicts the future position and velocity of both debris and satellites in the low zone of the orbit and identifies the conjunction possibilities by evaluating relative distances. [6][7]

For each debris object, relative distance is evaluated over a configurable time window:

$$d(t) = |r_s(t) - r_d(t)|$$

Coarse TCA detection is performed through discrete sampling. A quadratic polynomial is fitted to in the vicinity of the minimum:

$$f(t) = at^2 + bt + c$$

The refined TCA is obtained analytically:

$$t\{TCA\} = -\frac{\{b\}}{\{2a\}}$$

This approach improves temporal accuracy beyond coarse timestep resolution.

II.2 Collision Probability Modelling

Relative position at TCA is transformed into the RTN frame. [2][3] Even if the objects come close approach it is not deterministic due to uncertainty, so we compute probability of collision. (P_c)

A diagonal covariance matrix is assumed:

$$\sigma = \text{diag}(\sigma_r^2, \sigma_t^2, \sigma_n^2)$$

Collision probability is approximated by integrating the three-dimensional Gaussian distribution over a sphere defined by the combined hard-body radius:

$$P_c \approx \int V \frac{1}{(2\pi)^{\frac{3}{2}} \sqrt{|\Sigma|}} \exp\left(-\frac{1}{2} x^T \Sigma^{-1} x\right) dx$$

Monte Carlo validation is conducted to assess numerical stability [4].

CRITICAL Synthetic Encounter

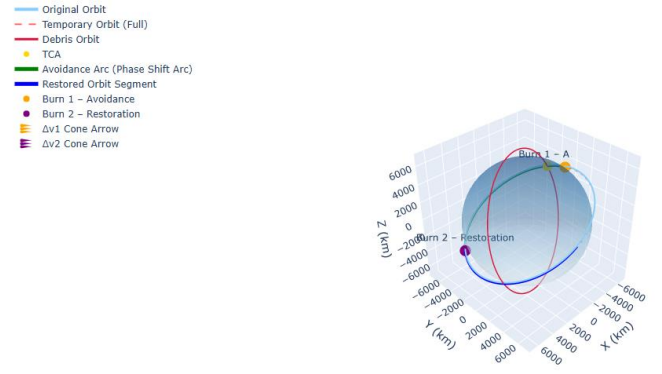


Figure 3 Critical Synthetic Encounter

Synthetic LEO Collision Scenario: STRONG_MANEUVER (Interactive)

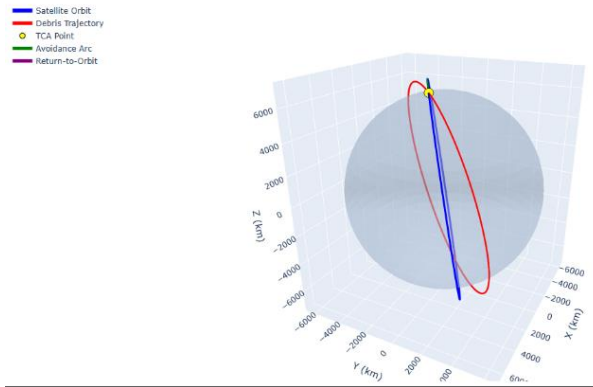


Figure 4 Synthetic LEO collision Scene for strong scenario

II.3 Maneuver Timing Optimization

Once collision risk is detected the system must decide the trajectory for maneuver and amount of ΔV supplied. The minimum distance and burn should be accurate with relative distances, velocity and time of conjunction. [7][9] [5]

Burn timing is defined as:

$$t_{burn} = t_{TCA} - \Delta t$$

Candidates lead times are swept over a defined range. For each candidate burn, resulting post-maneuver separation is evaluated [3]. The optimal burn minimizes ΔV subject to:

$$C = d_{miss} + \omega \cdot \frac{\Delta V}{C}$$

III. Physics-based approaches

Physics-based collision estimation methods originated from first formulations using the poisson distribution and ideas related to kinetic theory of gases. Today's physics-based collision estimations are formulated primarily using positional Gaussian distributions and covariance obtained from orbit determination. In general, they work by mapping the uncertainty of each relevant space object in an encounter

The main differences between classical physics-based collision estimation methods lie in their assumptions regarding shape of objects involved (spherical vs. Irregular); linearity of relative motion (short term encounters vs. Long-term encounters); modelling of uncertainty (Gaussian vs. Non-Gaussian); additional simplifications; etc. Most classical methods were designed specifically for short-term encounters and assume spherical shapes and Gaussian uncertainties.

Machine learning uses mathematical models and algorithms to examine data; identify patterns within data; make predictions/decisions without being explicitly programmed to do so for every task

Machine learning (ML) methods allow us to capture and utilize non-linear relationships between features present within our input data, automate the process of creating representations of the input data (representation learning), allowing us to avoid the manual creation of feature extraction algorithms, and offer a very high degree of capacity in terms of processing power. As a result, we see machine learning (ML) techniques being used increasingly in space operation due to these capabilities; however, they still have a significant amount of work to do to become operationally ready and gain the necessary level of trust.

anomalies that may require further evaluation and/or operator review and thus assist operators in managing increasing levels of activity in orbital environments.

The diagram illustrates the evolution of AI/ML models across three phases, each represented by a box with a title and a list of key components. Arrows indicate the progression from Phase-I to Phase-II, and from Phase-II to Phase-III.

Phase-I Data (Short-Term)	Phase-II Trust (Mid-Term)	Phase-III Hybrid (Mid-to-Long Term)
DATA • Data Preprocessing • Feature Engineering • Physics-Consistent Augmentation	TRUST • Explainable AI • Feature Attribution • Decision Transparency	HYBRID • Physics-Informed ML • Hybrid Architecture • Generative Models
Short-Term	Mid-Term	Mid-to-Long Term

The first thing to know about a regression model is that it has been trained on synthetic encounter data that includes the following variables:

- The target of the regression model was the analytically calculated required delta-v. The model's effectiveness in predicting required delta-v were measured by mean absolute error (MAE) and root mean squared error (RMSE).



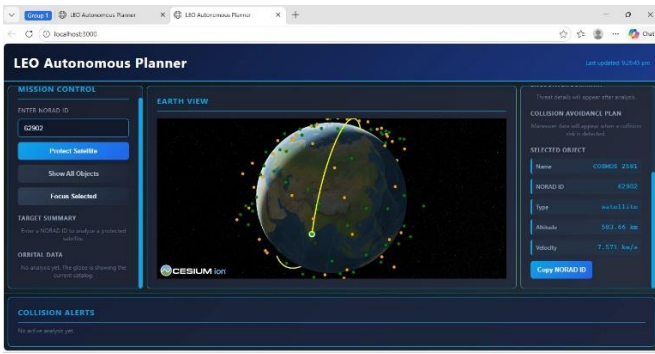


Figure 7 webpage for visualization of debris

V. Key challenges to operational deployment of ML-based approaches

Given the advancements in both the space and technological domains, it is natural to expect that elements of ML will eventually be integrated into operational systems. While physics-based methods have historically been the dominant paradigm for estimating PC when developing collision avoidance strategies, researchers are actively working toward integrating elements of ML. There are several factors contributing to the challenges of implementing ML in real-time environments. One key factor is the absence of suitable training datasets. There are very few publicly available datasets, and those that exist are severely imbalanced due to the fact that most collisions occur during low-risk situations. In addition to validating a ML model based on synthetic data, a ML model should be tested on a real-world dataset so that the model is able to represent the operational environment for which the model will be deployed.

Another significant challenge facing ML is validation. ML models will need to undergo rigorous evaluation prior to deployment in safety critical applications. This is because there are currently no standardized frameworks for evaluating and benchmarking the performance of ML models. In order for AI/ML approaches to be implemented in an operational capacity, missed detection rates need to be near zero. Operational personnel need confidence that conjunctions identified as low risk by the system will remain at that level and will not later evolve into potentially hazardous encounters. According to Ryan et al. (2022), none of the current ML approaches have achieved low enough missed detection rates for dismissing large numbers of conjunctions.

Another fundamental issue is that the process of miss distance evolution is intrinsically random. Ryan et al. (2022) observed that while covariance growth can generally be modelled accurately, miss distance fluctuation appears random and is therefore difficult to estimate. If little or no signal is present in miss-distance behaviour, there will exist an inherent limitation to AI/ML methodology [18]. Therefore, collision avoidance could become a limited application for AI/ML techniques due to insufficient predictive signal existing in miss distance behaviour,

although covariance evolution models may be modelled much better. To enable safe use of such methodologies for PC, further standardization and validation procedures will need to occur.

VI. Results

Results demonstrate:

- Improved TCA accuracy in less than one second over coarse sampling.
- Nonlinear relationship of collision probability as a function of covariance magnitude.
- Reduced ΔV through optimization of burn time.
- Near linear increase in run-time as a function of debris population size through use of multiprocessing.
- High correlation between regression model predictions and actual analytical ΔV .

VII. Discussion

A balance was achieved by the proposed framework between deterministic physics-based modeling and scalable computation. Use of regression-based estimation of ΔV minimizes computational overhead in high density screening environments [8][7][10]. The use of diagonal covariance assumed simplifies computation; however, it is limiting to the representation of correlated uncertainty. There is considerable opportunity for development with AI/ML (and hybrid) methodology that goes beyond PC estimation into all areas of the collision avoidance pipeline. For methodology based upon predictive PC estimation, the primary goal should be to reduce false negative responses and to produce reliable and understandable results. Predictive PC methodology will have a different role as the hard body radius characterization improves, and as maneuver commitment times move forward to include later times. Since uncertainty naturally decreases near the Time to Collision (TTC), many conjunctions self resolve without mitigation. Thus, unless ML and hybrid methodologies demonstrate superior performance to physics-based methodologies under lower-uncertainty conditions, these emerging operational realities must be taken into account when developing future ML-based collision risk frameworks [21][20].

Key areas where additional effort will be required in order to take this to the next level are. Explainability of how an AI/ML method works to increase the trustworthiness of those AI/ML methods. That is, why did it decide what you decided? Why was that a good idea? The results must be fully explainable; the key characteristics affecting the outcome must also be identified so that we can understand the reasoning behind any recommendation made by the AI/ML system.

The authors believe there should be greater emphasis on the data analysis, pre-processing of the data, data discovery, and identification of key characteristics from the dataset to extract as much value as possible from the existing small amount of data. The most significant data points within the dataset should be identified; many other types of orbit and environment parameters must also be included in the dataset. Additionally, proper training datasets should be used to allow maximum generalization [12][22][24].

Additional work with using a variety of techniques for hyperparameter optimization needs to be completed since most ML-based PC estimation methods now manually select hyperparameters. Well-designed architectures from scratch could potentially be created to enable us to explore the vast search space of deep architectures to discover models that are capable of capturing the inherent relationships in our training data. [11][17][21].

VIII. Conclusion

This paper presents a scalable combination screening and maneuvering optimization framework for Low Earth Orbit (LEO) Satellites. This framework incorporates advanced TCA estimation and three-dimensional collision probability modeling based on covariance, as well as machine learning assisted Delta-V estimation. Performance testing indicates improved temporal accuracy and scalability in the screening process. Simplifying certain aspects limits the operational relevance, however this architecture provides a solid basis upon which research grade and startup collision avoidance/mitigation systems may be developed.

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Abbreviations

- ML - Machine learning
- TCA - Time of Closest Approach
- PC - Collision Probability
- RTN - Return to Normal
- CA - Collision Avoidance
- CDM – Conjunction Data Messages